Graph Engineering

General Agents, Chapter 10/12 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/graph-engineering

EN: 'The agent forgets, the graph does not.' A single loop's memory (progress.md) only works for that one loop. When many loops/agents must share facts, they need typed, connected records -- nodes and edges.
Roman Urdu: 'Agent bhool jata, graph nahi.' Ek loop ki memory (progress.md) sirf usi loop ke liye kaam karti. Jab kai loops/agents facts share karein, unhein typed, connected records chahiye -- nodes aur edges.

1. Two Graphs, Different Jobs

- **Commit DAG** (remembers WHAT WAS TRIED -- nodes=commits, edges=parent_of) vs **Knowledge Graph** (remembers WHAT'S KNOWN -- nodes=entities/claims, edges=supports/about). They connect, but never merge.
- Skip graphs entirely when tasks are independent, answers come from one document, relations are fixed/simple, or provenance doesn't matter.

2. Extraction + Resolution + Provenance

- Extraction: structured-output prompts, a cheap model does volume work. Resolution: a STRONGER model proposes canonical clusters for the same entity under different names -- a false merge is the worst failure.
- **4 Invariants:** every claim has a source (or is marked 'inference'), every artifact has authoring run+version, every evaluation names its rubric, every superseded object stays addressable (never deleted).

3. Grounded Checker + Graph of Loops

- A grounded checker verifies claims against actual graph EDGES, not just impressions -- 'triple not found: these 2 edges are needed' beats 'seems off'.
- **Perez's 4 governance failures:** Gaming (Goodhart's law), Blindness upward, Conflict, Measurement decay. Fix with Anchors (unquestionable real measurements) and Frozen nodes (never auto-updated test sets).

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Commit DAG	Nodes=commits, edges=parent_of -- remembers what was tried.	Nodes=commits, edges=parent_of -- kya try hua yaad rakhta.
Knowledge Graph	Nodes=entities/claims, edges=supports/about -- remembers what's known.	Nodes=entities/claims, edges=supports/about -- kya pata hai yaad rakhta.
Resolution	Merging different names for the same entity -- a false merge is the worst failure.	Same entity ke alag naamon ko merge karna -- false merge sabse bari failure.
Grounded checker	Verifies a claim against real graph edges, not just an impression.	Claim ko real graph edges ke against verify karta, sirf impression nahi.
Anchors / Frozen nodes	Unquestionable real measurements / never-auto-updated test sets that keep a graph honest.	Unquestionable real measurements / kabhi-auto-update-na-hone-wale test sets jo graph ko honest rakhte.

Quick Self-Check

Question	Answer
Why do the Commit DAG and Knowledge Graph never merge?	They track fundamentally different truths -- commits are facts by construction, claims are evidence-backed statements that can be wrong.
Why is a false merge during resolution the worst failure mode?	It silently combines two different entities' facts under one identity, corrupting everything downstream that relies on it.
What makes a checker 'grounded' rather than just impressionistic?	It checks a claim against specific graph edges and can say exactly which edge is missing, instead of vaguely saying something 'seems off'.
What are Anchors and Frozen nodes for?	They keep a self-referential graph honest -- anchors are real measurements no loop can argue with, frozen nodes are test sets optimizing loops can never change.

Where This Leads / Sources

Leads to: Trusting the Checker (measuring checker quality itself). Full 8 practice-project graph builds in 08; 18-Q quiz in 10-test-your-understanding.

Source: docs/graph-engineering/ (this repo), content originally via Zia Tutor AI connector.